



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Modern Video Compression Standards

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Multimedia Communications



Outline

Introduction

H.264/AVC

- New modes and tools

- Profiles, levels, performance

- Network abstraction layer

H.265/HEVC

- Definitions

- Coding tools

- Stream

Latest standards

- Video Coding Standards: Usage and Market Share

- H.266/VVC

- AV1



Outline

Introduction

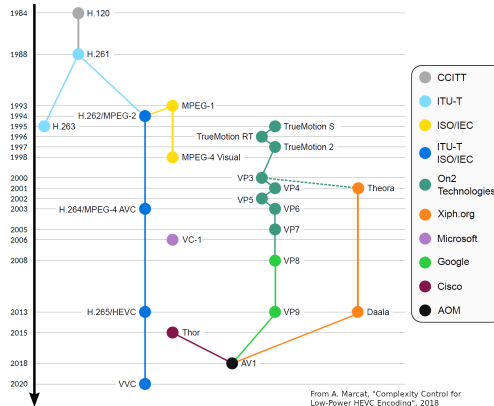
H.264/AVC

H.265/HEVC

Latest standards



Video standards: time-line





Video standards

Standardization bodies:

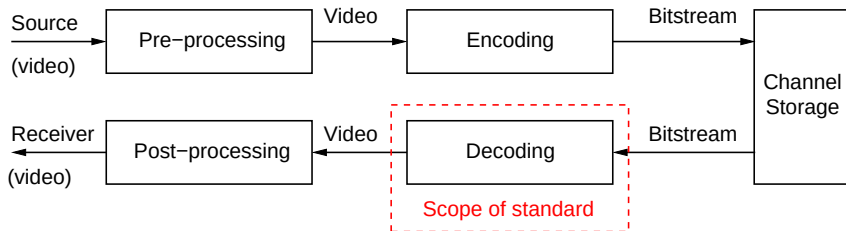
- ISO International Standardization Organization
- IEC International Electrotechnical Commission
- ITU International Telecommunication Union

Working groups

- ▶ MPEG (1988): ISO/IEC Moving Picture Expert Group
- ▶ VCEG (1997): ITU Video Coding Expert Group
- ▶ Those 2 groups have united into the Joint Video Expert Team (JVET)
- ▶ JVET has standardized H.264, H.265 and H.266



Scope of the standard



- ▶ The standard only defines the bitstream syntax through the **decoder**
- ▶ Goal: interoperability, competition
- ▶ However, it is easier to describe the encoding process



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- Profiles, levels, performance
- Network abstraction layer

H.265/HEVC

Latest standards



The H.264/AVC standard

- ▶ Developed in 1998-2003
- ▶ Standard approved as ITU-T Recommendation H.264 and as ISO/IEC International Standard 14496-10 (MPEG-4 part 10) Advanced Video Coding (AVC)
- ▶ **Hybrid coder**
- ▶ Only the video part has been standardized

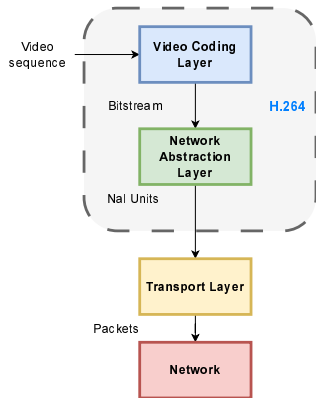


Goals

- ▶ Rate-distortion performance improvement
- ▶ Up to 60% of rate reduction for the same quality with respect to MPEG-2
- ▶ Applications:
 - ▶ Broadcast (cable, satellite, xDSL, DVB, etc.)
 - ▶ Storage on memory devices (optic, magnetic, *solid state*, etc.)
 - ▶ *Conversational services* over Ethernet, LAN, WiFi, xDSL, etc.
 - ▶ Video-on-demand and multimedia streaming over Ethernet, LAN, WiFi, xDSL, etc.
- ▶ Integration with the transport level taken into account



Structure of H.264



- ▶ Compression efficiency and integration with transport protocols impacts on global performance
- ▶ Therefore H.264 is organized into two conceptual (*layers*)
 - ▶ The Video Coding Layer (VCL) encodes the input video sequence into a bitstream
 - ▶ The Network Adaption Layer (NAL) creates the NAL units (packets), which are easy to manage for transmission or storage



Input format

- ▶ Input: digital YCbCr video
- ▶ In the standard, the Y component is referred as *Luma* and the Cb and Cr components as *Chroma*
- ▶ Default Chroma Sampling is 4 : 2 : 0 but Standard extensions allow for different color sampling scheme (Luma only, 4 : 0 : 0 and full resolution, 4 : 4 : 4)
- ▶ Encoding workflow: choose a prediction mode for block f_k ; compute de predictor $\hat{f}_k$; encode the prediction error e_k ;



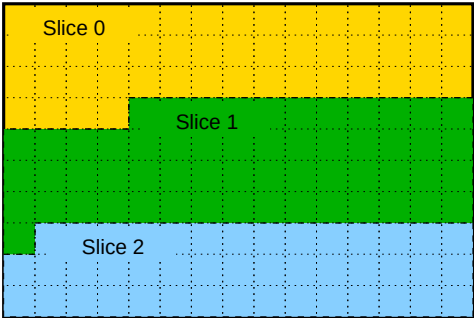
The macroblock

- ▶ **Macroblock:** a block of 16×16 Luma samples and two blocks of 8×8 Chroma samples
- ▶ The MB is the *coding unity*:
 - ▶ Each MB is coded with a *mode* (Intra, Inter, Skip)
 - ▶ Each mode produces: a prediction, a residual, some parameters
 - ▶ The prediction is built using
 - ▶ information available to the decoder
 - ▶ some parameters
 - ▶ The difference between the MB and the prediction is called residual
 - ▶ The residual and the prediction parameters are encoded



The slices

- ▶ In H.264/AVC the macroblocks are grouped into *slices*
- ▶ A slice is a set of macroblocks in *raster scan order*
- ▶ Other slice structures are possible
- ▶ However, most of the times a slices corresponds to an image





The slices

- ▶ Typically, a slice corresponds to an image
- ▶ Five types of slices exist:
 - ▶ Intra (I): All MB coded as Intra
 - ▶ Predictive (P): I + predictive modes with one reference* image
 - ▶ Bidirectional (B): P + predictive modes with two reference* image
 - ▶ Switch-Intra (SI) and Switch-Predictive (SP): for stream switch

* However, the **reference image can be different from a MB to the other**: therefore there is one **list** of references for P slices and two for B.



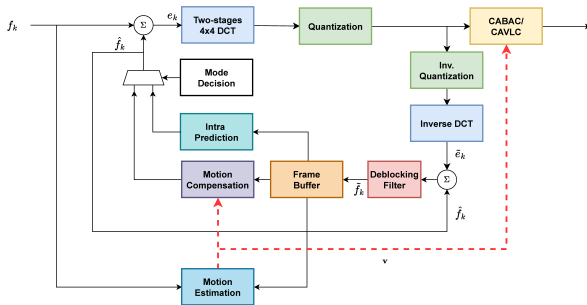
Pictures and Sequences

- ▶ An H.264 encoded video consists in a *sequence* of images, referred to as *coded pictures*
- ▶ A coded picture is made up of one or more slices
- ▶ A slice is divided into Macroblocks
- ▶ Each macroblock can be encoded using different *modes*, including modes that split the macroblock into sub-units
- ▶ For each unit, a different predictor can be used (but of the same type, i.e., Intra or Inter)
- ▶ The parameters of the predictor must be encoded, see later



H.264 video encoder

- ▶ New coding modes: Intra prediction, VBS, Skip
- ▶ New transform
- ▶ De-blocking filter
- ▶ New lossless coding (CABAC and CAVLC)





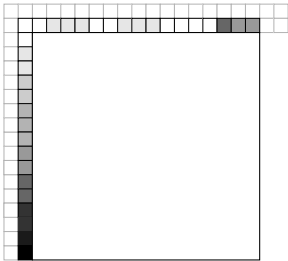
New coding modes

- ▶ *Intra* Modes
- ▶ *Inter* Modes
- ▶ *Skip* and *Direct* Modes



Intra modes with spatial prediction

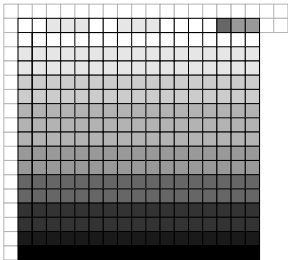
Prediction over 16x16 blocks: four possible predictors





Intra modes with spatial prediction

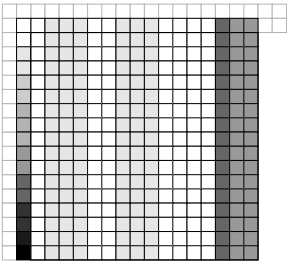
Prediction over 16x16 blocks: four possible predictors





Intra modes with spatial prediction

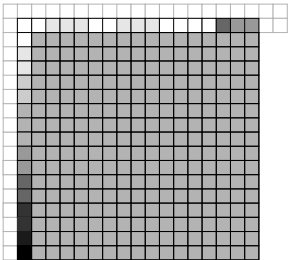
Prediction over 16x16 blocks: four possible predictors





Intra modes with spatial prediction

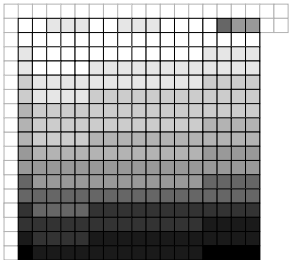
Prediction over 16x16 blocks: four possible predictors





Intra modes with spatial prediction

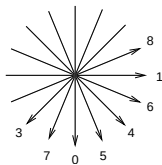
Prediction over 16x16 blocks: four possible predictors



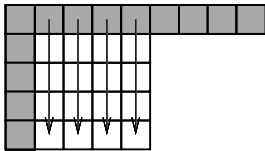


Intra modes with spatial prediction

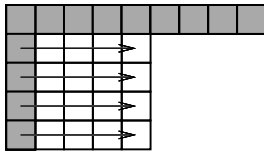
Prediction over 4x4 blocks (details), 9 prediction types: 8 directions and average



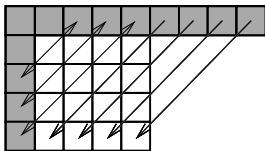
Directions



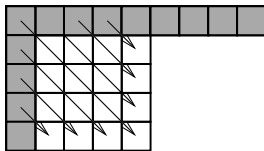
Mode 0



Mode 1



Mode 3

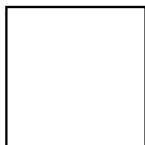


Mode 4

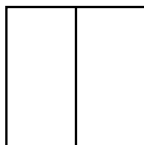


Inter modes with variable block-size

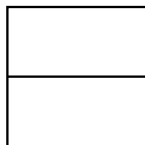
Each 16x16 block may be partitioned to have a finer representation of motion (a better prediction)



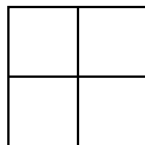
16x16



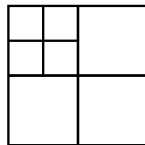
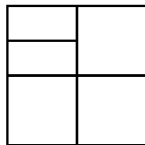
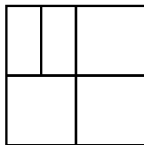
16x8



8x16



8x8





Motion estimation

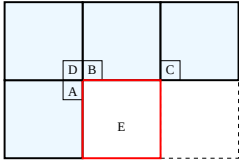
Variable block-size partition example





Motion representation*

- ▶ Quarter-pel motion vector
- ▶ Two successive interpolations: 6-taps filter and bi-linear filter
- ▶ Predictive vector coding median of available neighbors
- ▶ Vector may point outside the image





Motion compensation

- ▶ Very flexible image reference
- ▶ One list (P slices) or two lists (B slices) of reference images are available
- ▶ The encoder select the predictor as a MB of the images of the list(s)
- ▶ B prediction may combine two MBs with arbitrary coefficients



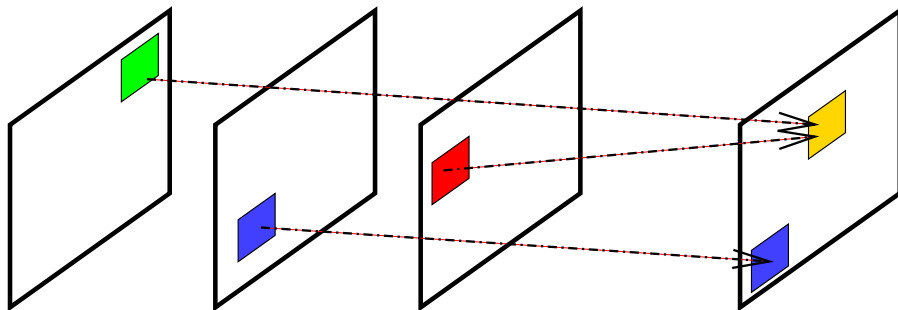
B-Slices

- ▶ Generalized B-slices
- ▶ Two reference image lists
- ▶ Multiple reference frame
- ▶ Flexible reference choice
- ▶ Flexible coding order
- ▶ Weighted average prediction



Motion compensation

Multiple references



References

Image to encode



Skip and Direct modes

- ▶ Skipped MB: it is a predictive MB but
 - ▶ zero bits are used to encode MV
 - ▶ zero bits are used to encode the residual
- ▶ The MV is estimated as the median of neighbors
- ▶ The prediction is not refined
- ▶ Coding cost: only the mode signaling
- ▶ Direct: Skip for B slices



Mode decision

- ▶ According to the slice type and on the profile, a subset of modes are available
 - ▶ For example, in Intra slices, only INTRA4 and INTRA16 modes are available
 - ▶ In P or B slices, also INTER16x16, INTER16x8 etc, SKIP/DIRECT are available
- ▶ The encoder decides which mode to use
- ▶ The mode decision process **is not standardized**
 - ▶ For example, one could choose the mode that minimizes $J = D + \lambda R$
 - ▶ The encoder can test all the available modes to evaluate the rate and the distortion of each one



Mode decision

- ▶ For each mode, there are some parameters to be transmitted:
 - ▶ The mode flag
 - ▶ For INTRA16, the prediction type (among 4)
 - ▶ For INTRA4, the prediction direction for each of the 16 sub-blocks
 - ▶ For INTER modes, all the motion vectors (from 0 to 16)
 - ▶ For DIRECT and SKIP, no other information is needed
- ▶ Then the prediction error is sent



New coding tools

- ▶ Deblocking filter
- ▶ New transform
- ▶ Lossless coders: CAVLC and CABAC
- ▶ Stream *switch* tools
- ▶ Robustness tools



Deblocking filter

Reference image



Current image



Motion-compensated image



Motion-compensated image has blocking artifacts corrected by residuals. At low rates, this correction may be insufficient.

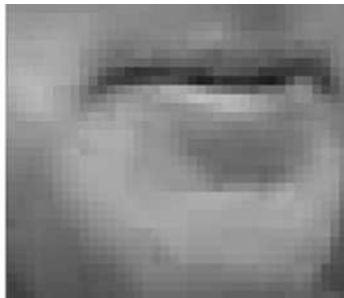
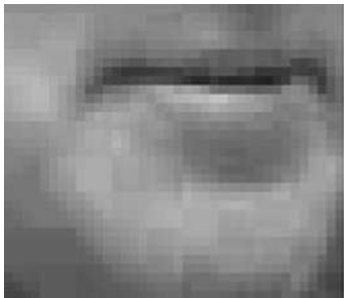


Deblocking filter

- ▶ Problem: blocking artifacts at low rate
- ▶ Cause: independent block coding
- ▶ Solution: post-filtering
 - ▶ does not affect coding (frame buffer)
 - ▶ Non normative, freedom
 - ▶ Needs additional memory
- ▶ Solution: in-loop filtering
 - ▶ Filtered images are inserted into the frame buffer and used as references
 - ▶ Therefore the filter must be normative
 - ▶ Improved objective and subjective performances
 - ▶ High complexity: branching, small blocks



Deblocking filter: example

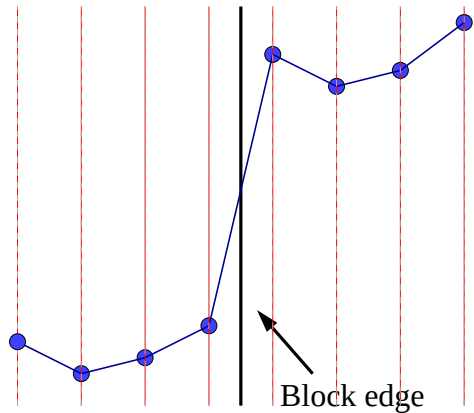


Peter List, Anthony Joch, Jani Lainema, Gisle Bjøntegaard, and Marta Karczewicz: "Adaptive Deblocking Filter", in *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 13, no. 7, July 2003



Deblocking filter

Discontinuity analysis





Deblocking filter*

- ▶ It reduces discontinuities between adjacent blocks 4×4
- ▶ Filtering is adapted to:
 - ▶ video sequence characteristics;
 - ▶ coding mode pf neighbors (stronger filter for Intra neighbors or neighbors using different references)
 - ▶ discontinuity amplitude and quantization step
- ▶ 4 types of filtering are possibles (in addition to the “non filtering” mode)
- ▶ Non-linear filter (low-pass and thresholding)



New transform

- ▶ 4x4 transform
 - ▶ The improved prediction reduces the spatial correlation of the residual
- ▶ Integer-coefficient DCT approximation
 - ▶ Encoder and decoder are always synchronized (no precision errors)
 - ▶ Coefficients : $0, \pm 1, \pm 2 \rightarrow$ implemented by bit-shift and sum or subtraction
- ▶ Two transform levels
 - ▶ 4x4 transform over the 16 DC coefficients of an Intra16 block
 - ▶ 2x2 transform over the 4 DC coefficients of Chroma blocks
- ▶ An 8x8 transform is possible (High profile) for blocks with high residual correlation



Coding order and quantization*

- ▶ In a MB first we send the 16 DC Luma coefficients (one per each 4x4 transformed block)
- ▶ Then, the other AC-Luma coefficients
- ▶ Finally, the Chroma coefficients
- ▶ Uniform quantization
- ▶ The quantization step depends on the quantization parameter QP
- ▶ The step is doubled when the QP is incremented by 6
- ▶ The rate increases of $\approx 12.5\%$ for a unitary increment of the QP



Lossless coding

- ▶ Two techniques are supported
 - ▶ A low-complexity method, based on variable length coding, with a dictionary depending on the context (CAVLC)
 - ▶ A high-complexity method, using context adaptive arithmetic coding (CABAC)
- ▶ Each improves remarkably w.r.t. previous standards
- ▶ Using probabilities that vary with the context is very effective



Context-adaptive variable length coding (CAVLC)*

- ▶ It is used in the *baseline* profile
- ▶ One first dictionary is used for residuals
- ▶ A second for any other syntax element (vectors, modes, signaling)
- ▶ Gain: 2–7% rate reduction w.r.t non-adaptive VLC



Context-adaptive binary arithmetic coder (CABAC)*

- ▶ Each syntax element is transformed in a binary format
- ▶ Each has its own arithmetic encoder
 - ▶ Adaptive: signal statistics are learned
 - ▶ Context-based: according to neighboring symbols the current one has different probabilities
- ▶ Complexity
- ▶ Gain 5–15% rate reduction w.r.t. CAVLC



Error robustness

- ▶ Mitigate error propagation
 - ▶ Unequal error protection (headers)
 - ▶ Limits imposed to predictions
- ▶ Allow synchronization recovery
 - ▶ Synchronization markers
 - ▶ *Marker emulation prevention*
- ▶ Limit error impact on visual quality
 - ▶ Recognize erroneous images
 - ▶ Error concealing

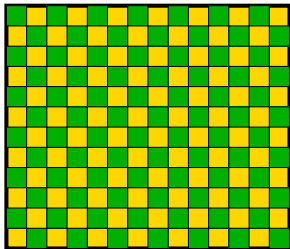
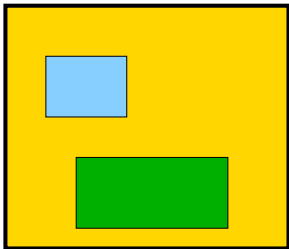


Error robustness in H.264*

- ▶ Flexible Macroblock Ordering
- ▶ Redundant Picture
- ▶ Data Partitioning
- ▶ SI and SP slices



Flexible macroblock ordering*



Slice group 0

Slice group 1

Slice group 2

- ▶ Independently encoded slices: a loss is limited to a slice
- ▶ *Unequal error protection* for Regions of Interest (ROIs)
- ▶ Improved *concealing* in the case of losses



Redundant picture

- ▶ In a *redundant picture* there can be one or more *redundant slices*
- ▶ A redundant slice is an additional encoded representation of an already encoded slice
- ▶ The already encoded version is called *primary slice*
- ▶ If the primary slice is received, the redundant slice is not used
- ▶ The redundant is only displayed if the primary is lost
- ▶ In order to reduce the rate overhead, redundant slices are typically strongly quantized (low quality)



Data partitioning*

- ▶ Data are partitioned into 3 groups
 1. Headers and motion vectors
 2. Residuals of Intra slices
 3. Residuals of other slices
- ▶ Then, UEP can be used on the different partitions
- ▶ Previous standards did not separate residuals (2 and 3)



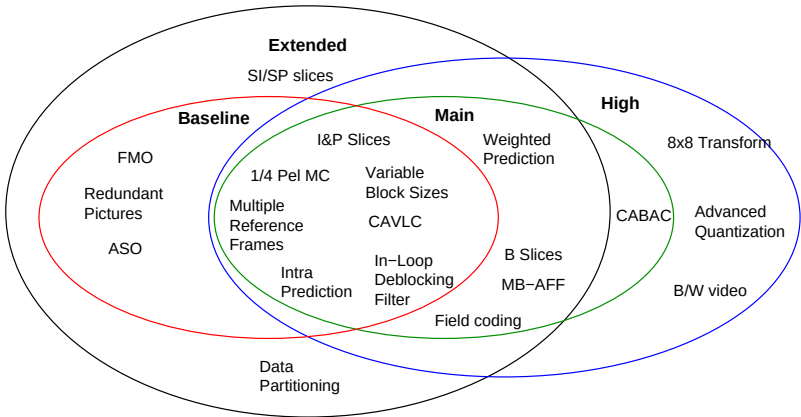
Profiles and levels*

It is possible to define the profiles and levels with a remarkable flexibility

- ▶ Seven profiles (Baseline, Extended, Main, High, High10, High4:2:2, High 4:4:4)
- ▶ 16 levels that can be combined with the profiles.



Profiles*





Profiles*

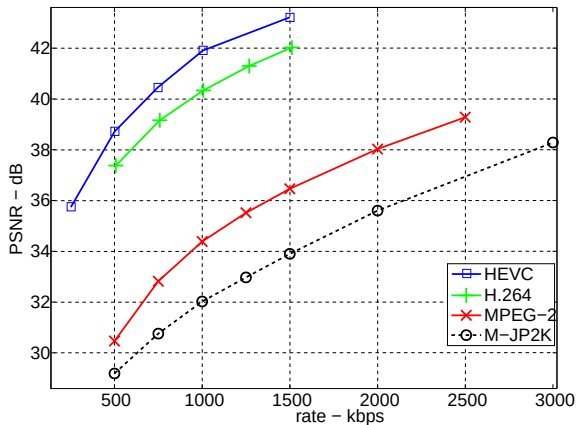
- ▶ Baseline Profile: low-cost applications (mobile devices, video-conference)
- ▶ Extended Profile: Streaming (robustness and switching tools)
- ▶ High Profile: the most popular profile for diffusion and the storage (TNT-HD, HD-DVD, BD), supersedes the Main Profile
- ▶ Other profiles target pre- and post-production and professional applications



Levels*

- ▶ Levels are limitations on parameters values
- ▶ This allows to limit the resources needed by the decoder
- ▶ Rate: 64 kbps to 240 Mbps for the Baseline and Extended profiles
- ▶ Rate: 80 kbps to 300 Mbps for the High Profile
- ▶ Resolutions: from QCIF, 15 fps to HD, 120 fps (or 4K, 30 fps)

Rate-distortion performance





H.264 – Decoded image examples



MPEG-2



H.264



H.264 – Decoded image examples



Motion JPEG2000



H.264



H.264 – Decoded image examples



MPEG-2



H.264



H.264 – Decoded image examples



Motion JPEG2000

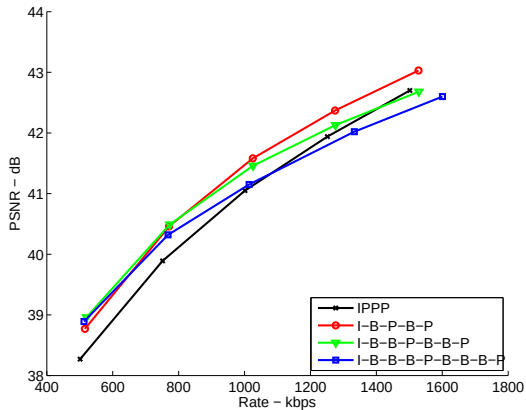


H.264



H.264 performance

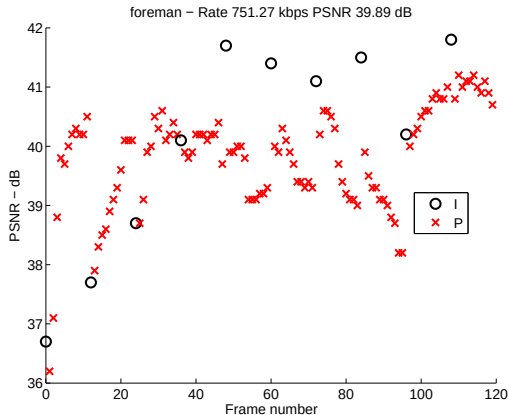
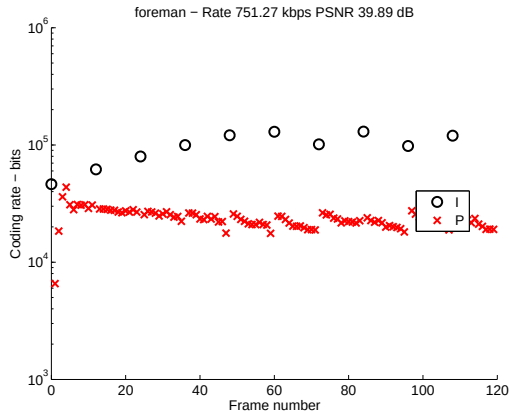
GOP structure effect





H.264 performance

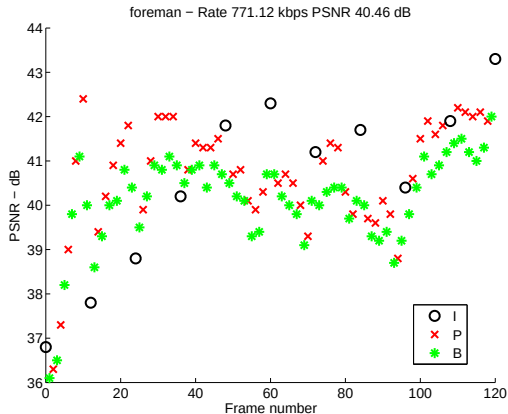
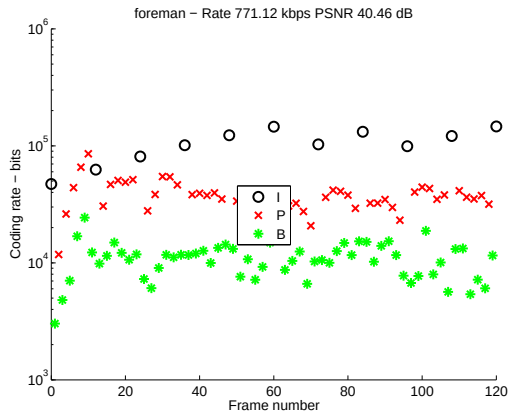
GOP IPPP: per-image rate and PSNR





H.264 performance

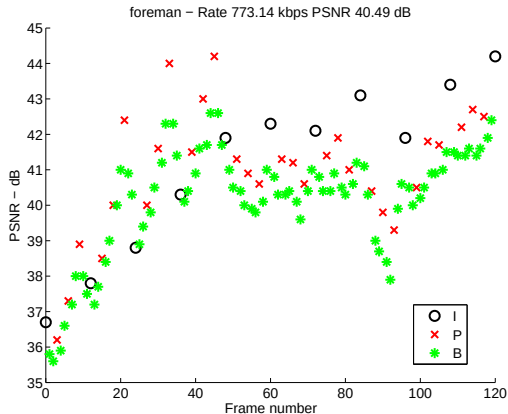
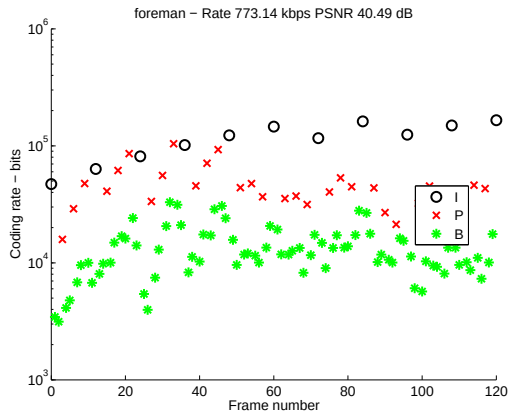
GOP IBPBP: per-image rate and PSNR





H.264 performance

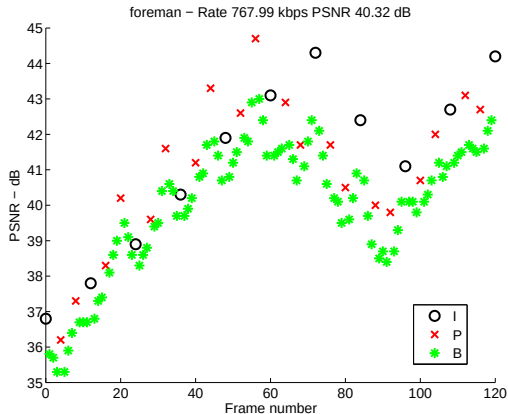
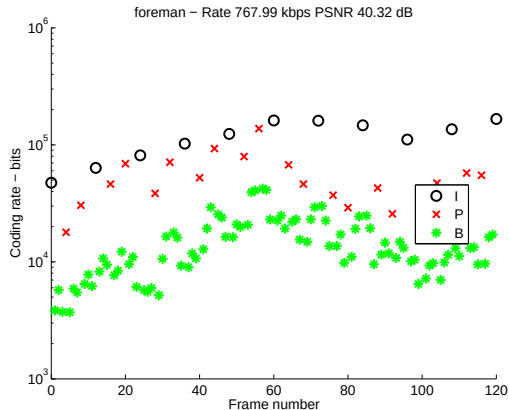
GOP IBBPBBP: per-image rate and PSNR





H.264 performance

GOP IBBBPBBBP: per-image rate and PSNR





Network Abstraction Layer

The NAL aims at providing “network friendliness” to H.264

- ▶ Simple interface between the VCL (video coding layer) and external systems
- ▶ Mapping over several transport layers:
 - ▶ RTP/IP for real time services
 - ▶ File format ISO MP4 for storage
 - ▶ MPEG-2 system for diffusion
 - ▶ H.32X for videoconferencing



NAL units*

- ▶ Encoded video is organized into NAL units (NALU)
- ▶ A NALU is a packet with a 1-byte header and $N - 1$ -bytes payload
- ▶ The header specifies the payload type
- ▶ If needed, payload contains some *emulation prevention bytes*
- ▶ *Byte-Stream* and *Packet-Transport* formats



Byte-Stream format NALU*

- ▶ When NALUs must be delivered as a ordered sequence of bytes
- ▶ Examples: MPEG-2 systems
- ▶ NALUs limits must be identifiable
- ▶ Solution: Start code prefix and emulation prevention bytes



Packet-Transport format NALU*

- ▶ Data are partitioned for the transport layer
- ▶ Examples: RTP/IP, UDP/IP
- ▶ In this case the start codes are not needed

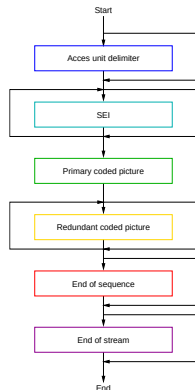


NALU types*

- ▶ VCL NALU (coding modes, motion vectors, quantized transform residual)
- ▶ Non-VCL NALU (header, parameter sets)
 - ▶ Parameter set per image and per sequence
- ▶ UEP can be used



Access unit



- ▶ An access unit (AU) allows to decode an image
- ▶ A *video sequence* is a sequence of AUs
- ▶ A sequence starts with an IDR (instantaneous decoding refresh) image
- ▶ A NALU stream may contain one or more *video sequences*



Outline

Introduction

H.264/AVC

H.265/HEVC

- Definitions
- Coding tools
- Stream

Latest standards



High Efficiency Video Coding

- ▶ Joint ISO/ITU-T standard
 - ▶ MPEG-H Part 2 (ISO/IEC 23008-2)
 - ▶ ITU-T H.265
- ▶ Goals: those of H.264, and in addition:
 - ▶ High and ultra-high definition
 - ▶ Parallel encoding
 - ▶ Improved R/D performance
- ▶ but it is still an hybrid codec with temporal and spatial prediction

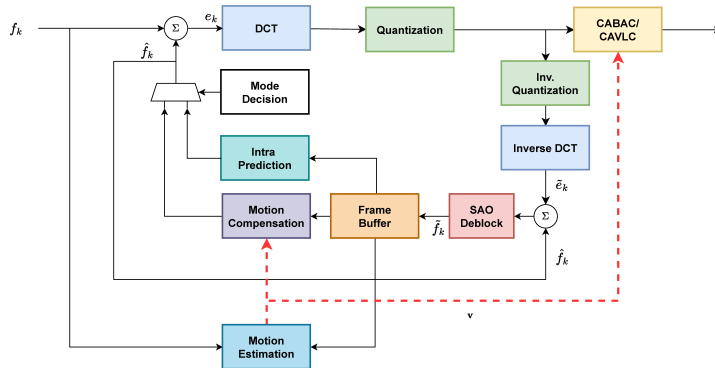


High Efficiency Video Coding

- ▶ *Call for Evidence*: April 2009
- ▶ *Joint Collaborative Team on Video Coding (JVT-VC)* : Jan. 2010
- ▶ *Call for Proposal* : Jan. 2010
- ▶ Subjective tests until April 2010
- ▶ Test Model under Consideration: April 2010
- ▶ HEVC Test Model v1 (HM 1.0): July 2010
- ▶ Final Draft International Standard: Jan. 2013
- ▶ HM 14.0 April 2014
- ▶ Several extensions are developed in parallel (Screen content, 3D, HDR, . . .)



HEVC: general scheme



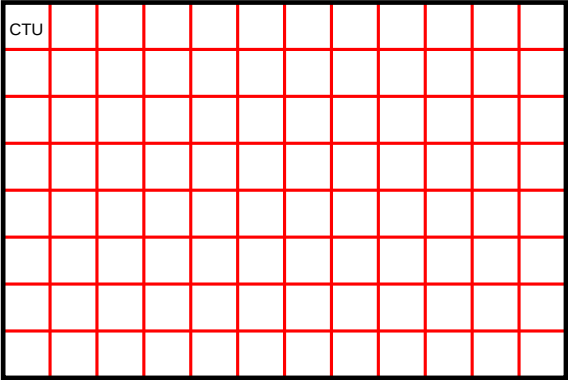


Data structures in HEVC

- ▶ Separation between units for encoding, prediction and the transform
- ▶ Units: set of data corresponding to the same rectangular region of the picture
- ▶ Block: a “unit” corresponding to one luminance and chrominance 2 sets of pixels
- ▶ The basic coding element is the *Coding Tree Unit* (CTU)
- ▶ Each CTU is divided into coding units (CU)
- ▶ Each CU is divided into prediction units (PU) and transform units (TU)



Coding tree units





Coding tree units

- ▶ Current picture is divided into CTU
- ▶ A CTU is formed by one luminance and 2 chroma Coding Tree Blocks (CTB)
- ▶ The chroma CTB have a smaller resolution, since in HEVC the input video is in 4 : 2 : 0 format
- ▶ The CTU size selected by the encoder among 16×16 , 32×32 and 64×64
- ▶ Larger CTUs give better RD performance but demand for higher complexity



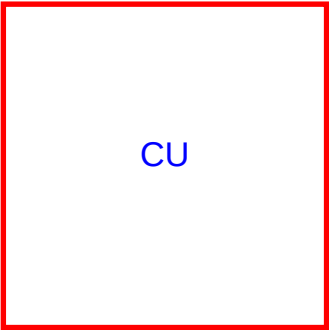
Coding units

- ▶ CTU are divided into coding units (CU)
- ▶ A CU is formed by one Luma and 2 chroma Coding Blocks (CB)
- ▶ CU always have squared shape
- ▶ The maximal CU size is that of the CTU (and for this reason CTUs are sometimes called LCUs, largest coding units)
- ▶ Minimum size is 8×8
- ▶ Quad-tree structure
- ▶ The coding mode decision (Intra or Inter) is done at the CU level

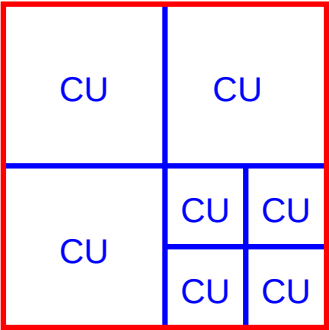


Coding units

CTU



CTU



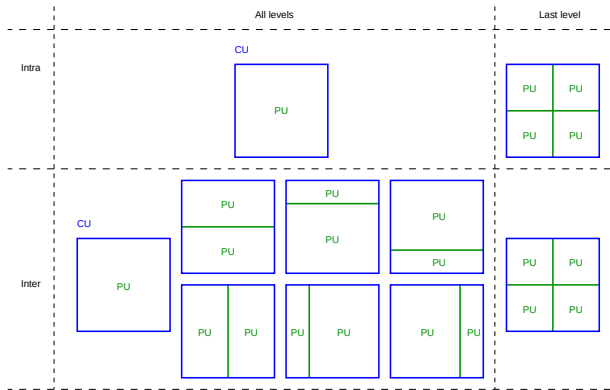


Prediction units*

- ▶ For Intra CU , the PU corresponds to the CU
 - ▶ Except for last-level (smallest) CUs which may be partitioned into 4 square PUs
- ▶ Inter CU may correspond to the PU...
- ▶ ... or may be partitioned into 2 PUs (six possible choices)
- ▶ or, – only for the last level – partitioned into 4 square PUs
- ▶ A PU is formed by one Luma and 2 chroma PBs
- ▶ For each PU the encoder writes into the stream the prediction information (prediction direction for Intra, motion information for Inter)
- ▶ the residual is transformed in the TU, which may have a different shape from the PU



Prediction units





Transform units*

- ▶ The CU may be divided into TU using a quad-tree scheme
- ▶ This is independent from the PU partition
- ▶ A TU may thus cover several PUs ...
- ▶ ... or cover just a portion of a PU
- ▶ Some information about TU partitioning are implicit by comparing maximum and minimum sizes of TU and CU

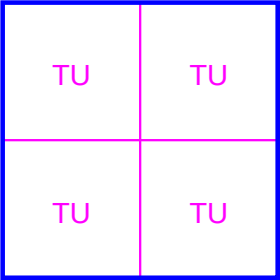


Transform units

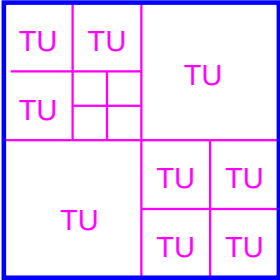
CU



CU



CU





Intra coding

- ▶ Intra prediction mode is chosen at the PB level
- ▶ The PB corresponds to the CB excepted for the last level
- ▶ 35 Intra coding mode are possible for luminance
- ▶ 33 directions + IntraDC + IntraPlanar
- ▶ It is independent from the block size (between 4×4 and 32×32)
- ▶ The prediction directions are denser near the horizontal than the vertical
- ▶ Prediction is computed with bilinear interpolation
- ▶ Reference values may be modified to reduce artifacts



Intra coding*

- ▶ The mode (direction) is encoded by creating a list of the most probable modes (MPM)
- ▶ An identical MPM list is created by the decoder
- ▶ Top and left neighbors are used if available and Intra-coded
- ▶ Unavailable neighbors are replaced by default values (planar, DC, vertical)
- ▶ A mode in the MPM list is coded with an index ("0", "1", "2")
- ▶ Otherwise, one of the remaining 32 modes is coded on 5 bits
- ▶ For the chroma PB it is only possible to choose among Planar, V, H, DC and "Direct" (the same as Luma)
- ▶ The chroma mode is directly encoded (without MPM)



Inter coding*

- ▶ The CB may correspond to the PB ...
- ▶ ... or may be partitioned into 2 PBs: horizontal high, middle, low; vertical left, center, right
- ▶ The CB cannot be partitioned into 4 PBs (it is more efficient to perform the partition at CB level)
- ▶ Only exception: the last CB level may be partitioned into 4 PBs



Inter coding

- ▶ As in H.264, Inter prediction may be performed using multiple references in a list
- ▶ B slices maintain two lists
- ▶ Motion estimation is performed at quarter-pixel precision
- ▶ Two different pixels are used to perform half-pel and quarter-pel interpolations



Inter coding

Merge

- ▶ The Merge mode allows sharing motion information among neighboring PBs
- ▶ It is similar to H.264 SKIP with 2 differences
 - ▶ Several candidates for MV are possible, and are arranged into a list; the MV index is explicitly encoded
 - ▶ The reference image index is also explicitly sent rather than deduced as in H.264



Inter coding*

Merge

- ▶ Shared motion information consists in
 - ▶ One or two motion vectors
 - ▶ One or two reference indexes
- ▶ A list of C candidates is created
- ▶ C is specified into the slice header
- ▶ If the MV that we want to encode is not in the list, we encode it by a prediction
- ▶ The prediction is selected between 2 candidates of the list



Inter coding*

Merge

- ▶ Merge candidates are, if available:
 - ▶ Five spatial candidates (neighboring PB's MVs)
 - ▶ One temporal candidate (bottom-right or co-localized PB of the reference image)
 - ▶ Generated candidates up to reaching 5 *different* candidates



Transform and entropy coding

- ▶ Four transform size are possible: 32×32 , 16×16 , 8×8 and 4×4
- ▶ Integer-coefficients separable transforms, approximating a DCT
- ▶ Smaller matrices are sub-sampled versions of larger ones
- ▶ For 4×4 blocks it is possible to use an alternative transform approximating DST
- ▶ DST reduces the rate of $\approx 1\%$ for 4×4 blocks in Intra mode
- ▶ HEVC features an improved and simplified version of CABAC



Deblocking filter*

- ▶ A new deblocking filter is implemented in order to reduce complexity
- ▶ It is applied only on a 8×8 grid
- ▶ Good trade-off between complexity and efficiency
- ▶ Only 3 filtering strengths are possible
- ▶ A sample adaptive offset (SAO) filter is applied after deblocking
- ▶ The SAO adds an offset as a function of the pixel value and of the region characteristics (flatness, minimum, contours, maximum)



High level syntax: slices

- ▶ The CTU can be grouped into slices or tiles
- ▶ The slices are formed by a CTU sequence in *raster scan order*
- ▶ As in H.264, slices are *self-contained*: Intra prediction cannot be performed across slice borders
- ▶ The slices are of type I, P and B
- ▶ The slices main target is to allow re-synchronization after data losses

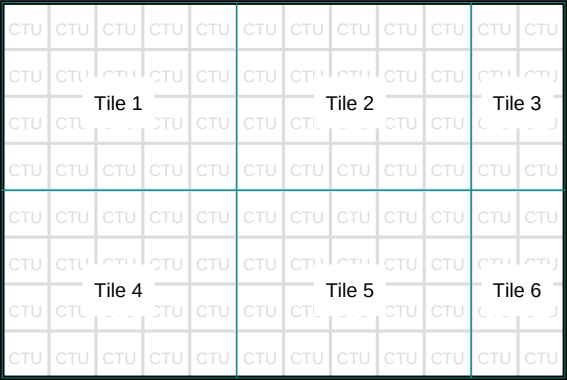


High level syntax: tiles

- ▶ The CTU may also be grouped into rectangular regions referred to as “tiles”
- ▶ The tiles are also “self-contained”
- ▶ The tiles are independent from slices
 - ▶ A slice may contain several tiles...
 - ▶ or a tile may contain several slices
- ▶ One of the tiles targets is to allow parallel encoding and decoding



Partition of a picture into tiles



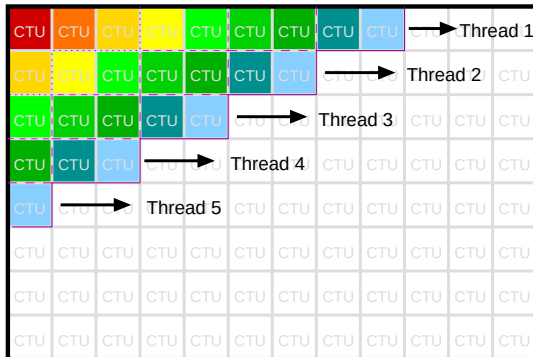


Wavefront processing

- ▶ A finer degree of parallelism may be achieved using the wavefront parallel processing (WPP)
- ▶ Slices are divided into rows of CTUs
- ▶ The encoding of a row may start as soon as two CTUs of the previous row have been encoded
- ▶ There is a delay of 2 CTUs between successive rows



Wavefront





Random access and bitstream splicing

- ▶ Decoding of a HEVC sequence may start at a random access points (RAP)
- ▶ A RAP may be
 - ▶ An IDR image (Instantaneous decoding refresh): all information after IDR in the bitstream can be decoded regardless past data
 - ▶ A CRA image (clean random access): it is followed in the bit-stream by past images which may be non-decodable
 - ▶ A BLA image (broken link access): The bitstream is cut at a CRA and it is spliced with another bitstream



Outline

Introduction

H.264/AVC

H.265/HEVC

Latest standards

Video Coding Standards: Usage and Market Share

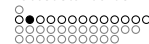
H.266/VVC

AV1



The H.266/VVC standard

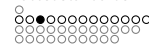
- ▶ Developed in 2015-2020
- ▶ Aims at –50% rate for the same quality of H.265/HEVC
- ▶ Targets high definitions, frame rates and wide color gamut
- ▶ Broader content: computer generated, videogames, immersive, 360
- ▶ Low-latency (gradual decoding refresh) and adaptive (Reference picture resampling) streaming
- ▶ Multilayer coding



VVC coding tools*

HEVC version 1	VVC version 1
Block partitioning	
64×64 max. CTU size	128×128 max. CTU size
Quadtree (QT)	Quadtree plus Multi-Type Tree (QT+MTT)
Coding Units (CU)	Coding Units (CUs)
Prediction Units (PU)	Chroma Separate Tree (CST)
Residual Quadtree	Local Dual Tree
Transform (RQT)	Virtual Pipeline Data Units (VDPUs)
Motion Compensated or Inter-Picture Prediction	
Merge Mode	Extended Merge Mode and MVP with History-based MV Prediction (HMVP)
Advanced MVP	Pair-wise Average MV Prediction Candidate Subblock-Based Temp. MV Pred. (SBTMVP)
	Merge with MVD (MMVD)
	Symmetric MVD (SMVD)
	Adaptive MV Resolution (AMVR)
8-tap IFs	8-tap Interpolation Filters (IF)
	Geometric Partitioning Mode (GPM)
	Bi-prediction with CU-level Weights (BCW)
	Combined Intra/Inter-picture Prediction (CIIP)
	Decoder-side MV Refinement (DMVR)
	Bi-Directional Optical Flow (BDOF)
	Affine Motion
	Pred. Refinement with Optical Flow (PROF)
Intra-Picture Prediction	
33 Angles	93 Angles
Linear interpolation	4-tap IFs (2 sets of filters)
DC, Planar	DC, Planar
	Position-Dependent Pred. Combination (PDPC)
	Multiple Reference Lines (MRL)
	Matrix-based Intra-picture Prediction (MIP)
	Cross-Component Linear Model (CCLM)
	Intra Sub-Partitions (ISP)
Transforms and Quantization	
Square transforms	Non-square transforms (max. 64×64)

From: Bross *et al.*, “Developments in International Video Coding Standardization After AVC, With an Overview of VVC”, Proceedings of IEEE 2020



VVC coding tools*

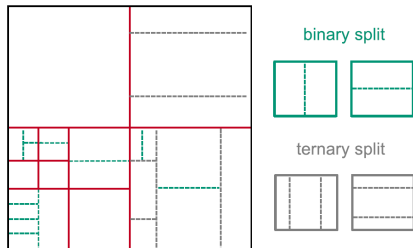
Entropy Coding	
CABAC	CABAC with high-accuracy multi-hypothesis probability estimates
Coefficient groups	Additional coefficient group sizes
Reverse diagonal, hor. and ver. coefficient scan	Reverse diagonal coefficient scan only Improved probability model selections for absolute transform coefficient levels
In-Loop Filtering	
Deblocking	Luma Mapping with Chroma Scaling (LMCS) Deblocking Boundary Handling Modifications Deblocking Long Filter
SAO	Luma-Adaptive Deblocking Sample Adaptive Offset (SAO) Adaptive Loop Filter (ALF) Cross-Component ALF (CC-ALF)
Special Modes	
PCM	
4×4 TS	4×4–32×32 Transform Skip (TS)
Trans. Quant. Bypass	
Quantization Bypass	
Screen Content Coding	
	Block-Level Differential PCM (BDPCM)
	Transform-Skip Residual Coding (TSRC)
	Intra-picture Block Copy (IBC)
	Palette Mode
	Adaptive Color Transform (ACT)
360-degree Video Coding	
	MV Wrap-Around
	Virtual Boundaries for in-loop filtering

From: Bross *et al.*, “Developments in International Video Coding Standardization After AVC, With an Overview of VVC”, Proceedings of IEEE 2020



VVC coding tools

Block partitioning



From: Bross *et al.*, "Developments in International Video Coding Standardization After AVC, With an Overview of VVC", Proceedings of IEEE 2020

- ▶ QT+MTT replaces quadtree with multiple unit types
- ▶ Larger CTUs (up to 128×128 pixels)
- ▶ CTU partitioned into QT
- ▶ QT leaves can be partitioned by a multitype tree (4 types) into CUs



VVC coding tools

Inter prediction

- ▶ Improved MV coding (several smart predictors)
- ▶ Improved Merge mode
- ▶ High-precision interpolation filter
- ▶ New tools: affine motion, non-rectangular split, bi-prediction with weights, motion refinement, inter/intra combination



VVC coding tools

Intra prediction

- ▶ VVC expands the ideas of HEVC Intra coding
- ▶ 65 directional plus 2 non-directional
- ▶ 28 directions are replaced by “wide angles” for rectangular blocks
- ▶ MPM with 6 candidates
- ▶ Other modes:
 - ▶ Position-dependent prediction combination
 - ▶ Multiple reference line
 - ▶ Matrix-based intra-picture prediction
 - ▶ Intra Sub-Partition



VVC coding tools*

Transform and Quantization

- ▶ Non-square transforms
- ▶ Multiple transform selection
- ▶ Low-frequency non-separable transform
- ▶ Sub-block transform
- ▶ Adaptive chroma QP offset, Dependent Quantization, and Joint coding of chroma residual



VVC coding tools*

In-loop filtering

- ▶ Deblocking filter based on the one in HEVC but extended with longer filters
- ▶ Luma-adaptive filtering mode designed specifically for HDR video
- ▶ SAO is the same as in HEVC
- ▶ Luma mapping with chroma scaling (LMCS) and adaptive loop filter ALF are new



VVC coding tools*

Other features

- ▶ Improved CABAC engine
- ▶ Screen content coding: skip prediction/transform, intra block copy, palette mode
- ▶ 360 video coding: signaling of sphere-to-plane mapping, MV wrap-around, virtual boundaries



VVC savings over HEVC and AVC

- ▶ Common test conditions: sequence resolution, frame-rate and content
- ▶ Four quantization points (QP 22, 27, 32, 37)
- ▶ Bjontegaard delta bit-rate (BD-rate): average rate increase for the same quality (PSNR or SSIM)
- ▶ The YUV PSNR is used:

$$\text{PSNR}_{\text{YUV}} = \frac{6\text{PSNR}_Y + \text{PSNR}_{C_B} + \text{PSNR}_{C_R}}{8}$$

- ▶ Three configurations: Random Access, Low-delay, All-Intra



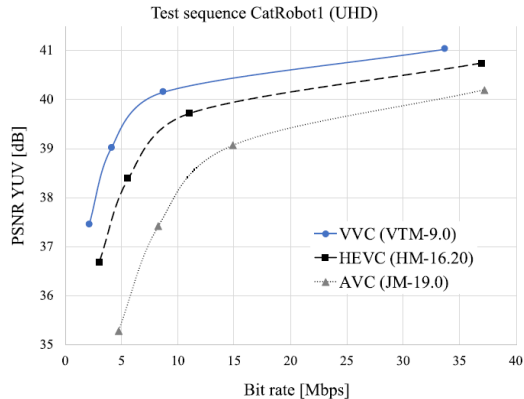
VVC savings over HEVC and AVC

Sequences Class	AVC High 10 Random Access	HEVC Main 10 Random Access	HEVC Main 10 Low Delay B	HEVC Main 10 All Intra
A1 (3840×2160)	71.6%	39.2%	—	30.4%
A2 (3840×2160)	69.8%	42.9%	—	28.6%
B (1920×1080)	65.0%	37.2%	31.8%	23.0%
C (832×480)	55.6%	30.3%	28.1%	22.3%
E (1280×720)	—	—	34.0%	25.7%
Average	64.8%	36.9%	31.1%	25.5%
D (416×240)	52.0%	27.5%	24.4%	17.5%
F (screen content)	64.9%	42.4%	43.1%	39.8%
F (screen content) Screen-Extended Main 10	—	26.4%	32.5%	17.8%

From: Bross *et al.*, "Developments in International Video Coding Standardization After AVC, With an Overview of VVC", Proceedings of IEEE 2020



VVC performance



From: Bross *et al.*, "Developments in International Video Coding Standardization After AVC, With an Overview of VVC", Proceedings of IEEE 2020



VVC performance

Random Access YUV BD-Rate Savings of VVC (VTM-9.0) Over VVC Without Specific Tool Sets

	Inter	Intra	Trans. & Quant.	Loop Filtering	All 4
A1	10.5%	7.7%	4.8%	5.5%	27.3%
A2	17.1%	2.4%	4.6%	10.1%	31.3%
B	11.5%	2.9%	4.8%	9.7%	26.2%
C	8.9%	3.0%	3.9%	5.5%	20.2%
Avg.	11.7%	3.8%	4.5%	7.8%	25.8%
EncT	199%	112%	127%	114%	513%
DecT	118%	98%	98%	108%	142%
D	9.5%	2.3%	3.8%	5.2%	19.4%
F	7.8%	4.5%	3.9%	4.8%	19.9%

From: Bross *et al.*, "Developments in International Video Coding Standardization After AVC, With an Overview of VVC", Proceedings of IEEE 2020

- ▶ VVC's coding efficiency improvement over HEVC stems from multiple coding feature
- ▶ The combined gains of all four tool sets are just slightly lower than the sum of the individual gains



VVC performance

Subjective results

MOS and PSNR-YUV BD-Rate Savings of VVC (VTM-10.0) Over HEVC (HM-16.22) and of an Optimized VVC Encoder (VVenC-0.1) Over VTM

UHD SDR Sequences	VTM vs. HM		VVenC vs. VTM	
	MOS	PSNR-YUV	MOS	PSNR-YUV
DrivingPOV3	61%	43%	11%	-12%
Marathon2	37%	33%	7%	-16%
MountainBay2	37%	39%	3%	0%
NeptuneFountain3	38%	27%	21%	-14%
TallBuildings2	41%	37%	16%	-5%
Average	43%	36%	12%	-9%

From: Bross *et al.*, "Developments in International Video Coding Standardization After AVC, With an Overview of VVC", Proceedings of IEEE 2020

- ▶ Subjective quality improvement is higher than objective (PSNR)
- ▶ Optimized encoders can achieve even better subjective performance, but with lower PSNR



AV1

- ▶ HEVC had a very complicated licensing system that hindered its diffusion
- ▶ AV1 was developed by the Alliance for Open Media
- ▶ Lead by Google, Microsoft, Cisco
- ▶ Supported by many platforms (Android, Chrome, Microsoft) and services (Youtube, Netflix, Vimeo, Bitmovin)



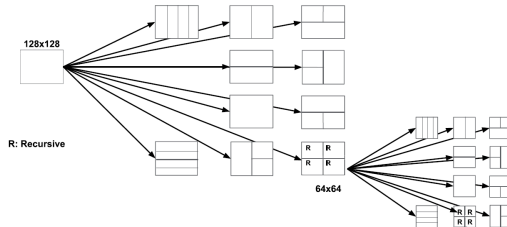
AV1

- ▶ AV1 supports 4:0:0 (monochrome), 4:2:0, 4:2:2, and 4:4:4 formats
- ▶ Bit-depths are 8, 10 or 12 bits
- ▶ Classical hybrid video codec: prediction, transform, quantization, VLC



AV1 Coding tools*

Hierarchical blocks, starting from a “superblock”

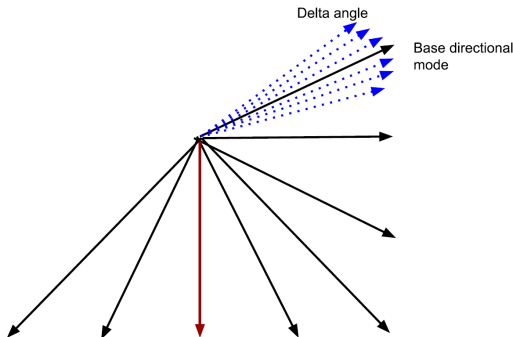


From: Han *et al.*, "Technical Overview of AV1", Proceedings of IEEE 2020



AV1 Coding tools*

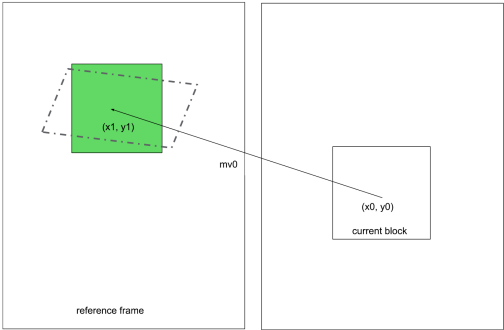
Intra coding is based on directional prediction with direction refinement Also, IBC and palette coding are available





AV1 Coding tools*

8-th pixel MC with 3 interpolation filters
Affine motion prediction

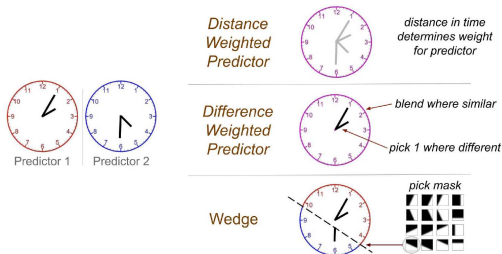


From: Han et al., "Technical Overview of AV1", Proceedings of IEEE 2020



AV1 Coding tools*

Compound prediction modes and overlapped MC

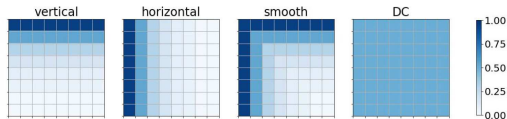


From: Han et al., "Technical Overview of AV1", Proceedings of IEEE 2020



AV1 Coding tools*

Compound of Inter and Intra modes



From: Han *et al.*, "Technical Overview of AV1", Proceedings of IEEE 2020



AV1 Coding tools*

Post processing

- ▶ Deblocking and directional filters
- ▶ Restoration filters
- ▶ Super-resolution
- ▶ Film grain synthesis



AV1 Coding tools*

Support to profile and levels

Profile	Bit-depth			Chroma sampling			
	8	10	12	4:0:0	4:2:0	4:2:2	4:4:4
Main	✓	✓		✓	✓		
High	✓	✓		✓	✓		✓
Professional	✓	✓	✓	✓	✓	✓	✓

From: Han *et al.*, "Technical Overview of AV1", Proceedings of IEEE 2020



AV1 performance*

Clip	Overall PSNR	SSIM	Enc Ins. Count	Dec Ins. Count
BalloonFestival_1280x720	-36.2	-43.0	88.2	2.5
CSGO_1080p	-29.2	-30.7	36.9	3.9
Campfire_1280x720	-31.8	-40.7	102.7	3.2
CatRobot_1280x720	-39.5	-39.4	37.3	3.0
DaylightRoad2_1280x720	-30.1	-29.8	42.5	3.1
Drums_1280x720	-37.0	-38.9	34.3	2.6
Market3Clipr2_1280x720	-38.4	-40.6	37.6	2.6
Netflix_Aerial_2048x1080	-31.4	-33.0	36.8	2.8
Netflix_Boat_2048x1080	-37.6	-38.6	33.2	2.4
Netflix_Crosswalk_2048x1080	-33.2	-35.3	37.7	2.5
Netflix_Dancers_1280x720	-38.2	-41.8	44.6	3.2
Netflix_DrivingPOV_2048x1080	-29.8	-30.1	43.2	2.9
Netflix_FoodMarket2_1280x720	-33.4	-35.0	35.8	2.5
Netflix_FoodMarket_2048x1080	-30.3	-33.0	43.2	3.1
Netflix_PierSeaside_2048x1080	-46.4	-42.1	38.3	2.9
Netflix_Square_2048x1080	-34.0	-34.8	29.6	2.8
Netflix_TunnelFlag_2048x1080	-39.3	-42.4	25.5	3.0
RollerCoaster_1280x720	-32.4	-34.9	35.8	3.2
ShowGirl2_1280x720	-28.6	-32.1	41.9	3.0
Tango_1280x720	-32.7	-32.7	36.8	3.3

Clip	Overall PSNR	SSIM	Enc Ins. Count	Dec Ins. Count
controlled_burn_1080p	-39.3	-42.6	41.2	2.7
crew_720p	-25.4	-28.2	44.3	3.7
crowd_run_1080p	-27.0	-30.8	31.4	2.8
dinner_1080p30	-37.5	-40.2	34.0	3.4
ducks_take_off_1080p	-33.1	-35.2	40.8	2.8
factory_1080p	-33.6	-40.6	29.6	3.2
in_to_tree_1080p	-26.7	-22.8	56.2	2.9
johnny_720p	-42.9	-43.8	43.9	2.4
kristenandsara_720p	-40.6	-38.4	44.0	2.8
night_720p	-34.8	-36.6	41.1	3.0
old_town_cross_720p	-38.4	-37.6	55.5	2.5
parkjoy_1080p	-27.2	-34.0	31.9	2.7
ped_1080p	-32.9	-38.2	35.8	3.3
red_kayak_1080p	-16.1	-14.7	37.5	3.5
riverbed_1080p	-17.4	-16.9	35.5	3.6
rush_field_1080p	-27.2	-30.0	34.5	3.2
rush_hour_1080p	-26.2	-34.5	47.9	3.7
shields_720p	-45.5	-44.5	44.8	2.8
station2_1080p	-56.4	-56.4	45.6	3.9



AV1 performance*

Resolution Group	A (UHD)				B (HD)				C (HD-DO)	
Codec	PSNR		VMAF		PSNR		VMAF		PSNR	VMAF
Sequence\BD-rate	AV1	VTM	AV1	VTM	AV1	VTM	AV1	VTM	AV1	AV1
AirAcrobatic	-12.1%	-25.5%	-12.0%	-28.6%	-2.6%	-21.7%	4.2%	-20.0%	13.3%	-0.1%
CatRobot	-6.2%	-38.0%	-12.8%	-39.6%	-4.0%	-37.7%	-10.4%	-41.2%	-2.1%	-11.3%
Myanmar	4.3%	-17.2%	1.3%	-21.3%	6.5%	-15.5%	3.5%	-18.6%	8.4%	5.1%
CalmingWater	-15.5%	-21.5%	-9.6%	-18.9%	-15.7%	-22.6%	-10.2%	-19.6%	-13.0%	-10.4%
ToddlerFountain	-6.6%	-18.7%	-2.0%	-17.4%	-8.1%	-18.2%	-3.7%	-16.4%	-7.8%	-7.3%
LampLeaves	-6.8%	-26.2%	-6.2%	-26.1%	-2.8%	-23.7%	-0.4%	-24.8%	6.7%	-1.2%
DaylightRoad	-3.8%	-38.0%	-12.4%	-40.3%	-0.9%	-37.6%	-10.4%	-42.4%	0.9%	-9.8%
RedRock	-3.5%	-32.5%	-9.0%	-37.9%	0.8%	-31.5%	-6.4%	-37.7%	16.1%	-8.0%
RollerCoaster	-15.3%	-39.9%	-14.5%	-41.7%	-7.9%	-38.9%	-11.5%	-39.8%	-6.6%	-13.5%
Average	-7.3%	-28.5%	-8.6%	-30.2%	-3.8%	-27.5%	-5.0%	-28.9%	1.8%	-6.3%

From: Zhang, Fan, et al. "Comparing VVC, HEVC and AV1 using objective and subjective assessments." arXiv preprint arXiv:2003.10282 (2020).



AV1 performance

- ▶ Overall, HM and AV1 are not significantly different in terms of perceived quality at the same bit rates and all resolution-groups.
- ▶ VVC showed significantly better performance than HM and AV1
- ▶ New (optimized) versions of VVC software are supposed to be even better



MPEG-2

- ▶ **Current Usage:**
 - ▶ Digital television broadcasting (DVB-T, ATSC)
 - ▶ DVD video
 - ▶ Some satellite and cable TV systems
- ▶ **Market Share:**
 - ▶ Global: Declining, but still significant in broadcasting
 - ▶ Europe: Still used in some DVB-T broadcasts
- ▶ **Notes:** Exact market share figures are difficult to determine due to the standard's age and varied use cases.

Sources:

- ▶ DVB Project: <https://dvb.org/standards/dvb-t2/>
- ▶ ATSC: <https://www.atsc.org/standards/atsc-1-0-standards/>



H.264/AVC

- ▶ **Current Usage:**
 - ▶ Streaming services (e.g., YouTube, Netflix)
 - ▶ Blu-ray discs
 - ▶ Digital TV broadcasting (older systems)
 - ▶ Videoconferencing
- ▶ **Market Share:**
 - ▶ Global: Approximately 82% of encoded internet video (2020)¹
 - ▶ Europe: Widely used, specific figures not available

¹Cisco Annual Internet Report (2018-2023)

<https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet->



H.265/HEVC

▶ Current Usage:

- ▶ 4K UHD streaming
- ▶ UHD Blu-ray discs
- ▶ Some digital TV broadcasting (e.g., DVB-T2)
<https://dvb.org/standards/dvb-t2/>

▶ Market Share:

- ▶ Global: Around 12% of encoded internet video (2020)²
- ▶ Europe: Growing, especially in 4K broadcasts and streaming

²Cisco Annual Internet Report (2018-2023)



H.266/VVC

- ▶ **Current Usage:**
 - ▶ Still in early adoption phase
 - ▶ Potential for 8K streaming and broadcasting
 - ▶ VR and 360-degree video applications
- ▶ **Market Share:**
 - ▶ Global: Not yet significant, adoption is just beginning
 - ▶ Europe: Early trials and testing phases
- ▶ **Notes:** As a new standard (finalized in 2020), widespread adoption is still pending.

Sources:

- ITU:
<https://www.itu.int/en/mediacentre/Pages/pr13-2020-New-video-coding-standard-VVC-MPEG-I-Part-3>



VP9

- ▶ **Current Usage:**
 - ▶ YouTube (for some 4K and HDR content)
 - ▶ Some streaming platforms (e.g., Netflix for specific devices)
 - ▶ WebM container for web videos
- ▶ **Market Share:**
 - ▶ Global: Estimated 3% of internet video traffic (2020)³
 - ▶ Europe: Used in web streaming, specific figures not available

Sources:

- ▶ Google Developer Blog: <https://developers.google.com/media/vp9>
- ▶ WebM Project: <https://www.webmproject.org/vp9/>

³Based on various industry reports, exact figures vary



AV1

- ▶ **Current Usage:**
 - ▶ YouTube (for some content)
 - ▶ Netflix (testing and gradual rollout)
 - ▶ Some streaming platforms and web browsers
- ▶ **Market Share:**
 - ▶ Global: Growing, but still less than 1% of internet video traffic (2021)⁴
 - ▶ Europe: Early adoption phase, specific figures not available
- ▶ **Notes:** Adoption is increasing but still limited by hardware support and encoding complexity.

Sources:

- ▶ Alliance for Open Media:
<https://aomedia.org/av1-beats-x264-and-libvpx-vp9-in-practical-use-case/>
- ▶ Netflix Tech Blog:
<https://netflixtechblog.com/netflix-now-streaming-av1-on-android-d5264a515202>



Additional Video Standards (1/2)

- ▶ **AVS (Audio Video Standard)**
 - ▶ Developed in China, used mainly for Chinese digital TV
 - ▶ AVS1 (2005), AVS2 (2016), AVS3 (2019)
- ▶ **MPEG-4 Part 2**
 - ▶ Predecessor to H.264/AVC
 - ▶ Initial release: 1999
 - ▶ Still used in some legacy applications (DVX)
- ▶ **VC-1**
 - ▶ Developed by Microsoft
 - ▶ Standardized in 2006
 - ▶ Used in HD DVD, some Blu-ray Discs, and Xbox 360
- ▶ **Dirac**
 - ▶ Developed by BBC
 - ▶ Initial release: 2008
 - ▶ Used primarily in broadcasting applications
- ▶ **JPEG XS**
 - ▶ Designed for professional video and broadcast



Additional Video Standards (2/2)

- ▶ **EVC (Essential Video Coding)**
 - ▶ Developed by MPEG
 - ▶ Finalized in 2020
 - ▶ Aims to offer an alternative with fewer licensing issues than HEVC
- ▶ **LCEVC (Low Complexity Enhancement Video Coding)**
 - ▶ MPEG standard to enhance efficiency of other codecs
 - ▶ Finalized in 2020
- ▶ **AV2**
 - ▶ Successor to AV1, currently in development
 - ▶ Expected release: Not yet announced
 - ▶ Aims to surpass VVC performance
- ▶ **VVC Screen Content Coding (VVC-SCC)**
 - ▶ Extension of VVC optimized for computer-generated content
 - ▶ Part of VVC standard, finalized in 2020

Note: Release dates refer to the first complete version or standardization, where applicable.



Conclusions

What's next?

- ▶ It may be difficult to significantly surpass VVC for quite a few years to come
- ▶ AI is promising but:
 - ▶ Difficult to implement at high speed
 - ▶ Difficult to implement at low cost
- ▶ Subjective quality based coding
- ▶ Video coding for machines, e.g., for self-driving vehicles
- ▶ Emerging formats (coding of point clouds, textures mapped onto moving 3-D meshes, and plenoptic light field coding) (come and see in the *Advanced Multimedia System* course)